Unit 1

Introduction to Project Management

1.1 Experiment 1: Project Requirement Gathering and Analysis

1.1.1 Software Project Management and Scope

Software project management encompasses the planning, organizing, monitoring, and controlling of software development projects [2]. The AI-Powered Sanskrit NLP Framework project scope includes the development of an NLP system capable of morphological analysis, semantic parsing, and knowledge extraction from classical Sanskrit texts using Paninian grammar.

1.1.2 Requirement Engineering and Elicitation

Requirement engineering involves the systematic process of eliciting, analyzing, documenting, and validating requirements [3]. The requirement gathering process for this project employed a hybrid approach combining three primary techniques.

Primary Technique: Facilitated Workshops

Three facilitated workshops were conducted with a duration of 2–3 hours each, involving:

- Two Sanskrit linguists specializing in Paninian grammar from Banaras Hindu University
- Two NLP researchers with experience in low-resource language processing
- The project manager and two senior NLP engineers

Secondary Technique: Document Analysis

The following documents were analyzed:

- Ashtadhyayi (Panini's grammar) – selected sutras relevant to morphological analysis [4]

- Existing Sanskrit NLP tools documentation (Sanskrit Heritage Engine [5], DCS, SanskritTagger)
- Project charter and problem statement

Tertiary Technique: Survey

A structured survey was distributed to 15 academic institutions across India, USA, and Europe working in digital humanities and computational linguistics. Response rate: 73% (11 responses).

1.1.3 Functional Requirements

Table 1.1: Functional Requirements for Sanskrit NLP Framework

ID	Requirement Description	Source	Priority
FR01	Morphological analysis of Sanskrit words using Paninian rules (Ashtadhyayi) [4]	Workshop	High
FR02	Semantic parsing of classical Sanskrit verses (subanta and tinanta structures)	Workshop	High
FR03	Knowledge extraction in subject-predicate-object triple format	Survey	High
FR04	REST API for researchers to submit Sanskrit text programmatically	Survey	Medium
FR05	Export parsed grammar trees as XML and JSON formats	Document analysis	Medium
FR06	Batch processing capability for large Sanskrit corpora (e.g., Mahabharata)	Workshop	High
FR07	Bilingual interface (Devanagari + IAST transliteration)	Survey	Medium

FR08	User authentication and role-based access for academic users	Document analysis	Low
FR09	Versioning of grammar rule updates for reproducibility	Workshop	Medium
FR10	Integration with existing Sanskrit digital libraries (GRETIL, SARIT)	Survey	Medium

1.1.4 Non-Functional Requirements

Table 1.2: Non-Functional Requirements

Category	**Requirement**	**Target Metric**	**Priority**
Performance	Morphological analysis latency	< 2 seconds per verse (up to 100 words)	High
Accuracy	Precision on test dataset	$> 92\%$ validated by Sanskrit experts	High
Scalability	Maximum corpus size	10 million word tokens	Medium
Availability	API uptime for academic use	99.5%	Medium
Security	Data protection and access control	Role-based access; no public modification	Low
Maintainability	Modular design for rule updates	New sutra addition within 2 person-days	High
Usability	Learning curve for researchers	< 4 hours for basic API usage	Medium
Reliability	Mean Time Between Failures (MTBF)	> 200 hours of continuous operation	Medium

1.1.5 Requirement Prioritization using MoSCoW Method

The MoSCoW prioritization framework [6] was applied to classify requirements:

Must Have (Critical for v1.0)

- Morphological analyzer based on Paninian rules (FR01)
- Panini rule engine with support for minimum 500 sutras
- Semantic parser for basic verse structures (FR02)
- Accuracy validation framework (NFR-Accuracy)

Should Have (Important but not blocking v1.0)

- Knowledge extraction module (FR03)
- REST API with basic documentation (FR04)
- Batch processing for corpora up to 1 million words (FR06)

Could Have (Desirable for v2.0)

- Visual syntax tree visualization
- Bilingual interface (FR07)
- Integration with external digital libraries (FR10)

Won't Have (Out of scope for this project)

- Speech synthesis or recognition for Sanskrit
- Machine translation from Sanskrit to English or Hindi
- Optical character recognition (OCR) for handwritten manuscripts

1.1.6 Requirements Traceability Matrix

Table 1.3: Requirements Traceability Matrix

Req ID	Source	Business Driver	Validation Method

FR01	Workshop (Linguists)	Digital preservation of Paninian grammar [4]	Unit test with 500 annotated Sanskrit words
FR02	Workshop (Linguists)	Semantic understanding for research	Expert review of 100 semantic frames
FR03	Survey (Researchers)	Structured knowledge extraction	Sample extraction from Ramayana corpus
FR04	Survey (Researchers)	Accessibility for external users	API endpoint testing with Postman
NFR-Perf	Feasibility study	Acceptable user experience	Load test with 50 concurrent requests
NFR-Acc	Workshop	Scholarly credibility	Cross-validation with 3 Sanskrit experts

1.1.7 Project Scope Management

In Scope

1. Development of morphological analyzer covering minimum 500 Panini sutras
2. Semantic parser for classical Sanskrit texts (pre-1000 CE)
3. Knowledge extraction module outputting structured triples
4. REST API with Swagger/OpenAPI documentation
5. Deployment on cloud infrastructure (AWS/GCP)
6. Sample Jupyter notebooks for researchers
7. User documentation and API reference

Out of Scope

1. Mobile application development
2. Real-time processing of streaming text
3. Support for post-1000 CE Sanskrit (Kavya, modern Sanskrit)
4. Integration with proprietary commercial systems

5. Training data annotation beyond project corpus

Acceptance Criteria

1. Morphological analysis accuracy $\geq 92\%$ on gold-standard test set [7]
2. API response time ≤ 2 seconds for 100-word input
3. Successful batch processing of Mahabharata (approximately 1.8 million words)
4. Positive validation from at least 2 external Sanskrit linguists

1.2 Experiment 2: Project Identification Process Methodology and Stakeholder Description

1.2.1 Project Identification and Selection

The project identification followed a structured five-phase methodology as documented in the PMBOK Guide [2] and Kerzner's project management framework [8]:

Table 1.4: Project Identification Phases

Phase	**Activity**	**Outcome for Sanskrit NLP Project**
Problem Identification	Systematic analysis of gaps in existing Sanskrit NLP tools [9]	Identified lack of integrated Paninian grammar engine with modern ML
Opportunity Analysis	Assessment of AI and NLP advancements for classical language processing	AI transformer models can be adapted for Sanskrit morphology
Feasibility Study	Technical, economic, and operational feasibility assessment [3]	Technical: Feasible with rule-ML hybrid; Economic: INR 33-41 Lakhs; Operational: High scholarly demand
Project Justification	Alignment with cultural, academic, and research objectives	Cultural preservation, linguistic research, AI for low-resource languages

Approval Gate	Submission to funding and oversight bodies	Approved by SRM, IIT (BHU) Sanskrit Department, and National Mission for Manuscripts

1.2.2 Feasibility Study Details

Technical Feasibility

- Existing Panini rule formalisms can be extended [5]
- Hybrid approach: rule-based morphological analysis combined with ML-based disambiguation [7]
- Technology stack: Python, spaCy, PyTorch, FastAPI
- **Identified Risk**: Complexity of sandhi (phonetic merging) rules requires expert linguist involvement

Economic Feasibility

- Total estimated cost: INR 33–41 Lakhs (using COCOMO model [10])
- Expected benefits: Academic licenses (INR 1.5 Lakhs per year per institution) and research grants
- Net Present Value (10% discount rate): Positive (INR 12.6 Lakhs over 5 years)
- Payback period: 3–4 years

Operational Feasibility

- Linguist availability: 3 senior Sanskrit experts willing to collaborate
- NLP engineering team: 6 engineers can be recruited within 2 months
- Infrastructure: Cloud services readily available
- **Constraint**: Limited number of Sanskrit NLP experts globally

1.2.3 Initiation Phases and Contracting

The project initiation involves the following phases:

1. **Project Charter Development**: Formal authorization document signed by sponsor

2. **Stakeholder Identification**: Mapping all parties with interest in project outcomes
3. **Contracting**: Memorandum of Understanding between SRM, IIT (BHU), and National Mission for Manuscripts

1.2.4 Stakeholder Analysis

Stakeholder analysis follows the framework proposed by Freeman [1] and Clegg et al. [11].

Table 1.5: Stakeholder Identification and Analysis

Stakeholder	**Role in Project**	**Interest Level**	**Power or Influence**	**Expectation**
UGC and MHRD (Sponsor)	Funding approval and oversight	High	High	On-time delivery with scholarly impact
Project Manager	Planning, execution, and reporting	High	Medium	Budget and schedule adherence
Sanskrit Linguists (3)	Grammar modeling and validation	Very High	High	Accurate Panini implementation [4]
NLP Engineers (4)	Model development and API	High	Medium	Clear requirements with modern stack
Data Scientists (2)	Corpus training and annotation	Medium	Medium	Quality annotated data
IGNCA (Cultural body)	Preservation partner	Medium	Medium	Usable digital archive
University Researchers	End users	High	Low	Accurate and easy-to-use API
Graduate Students	End users	Medium	Low	Learning resource and documentation

Cloud Provider (AWS)	Infrastructure hosting	Low	Low (critical)	Service level compliance

1.2.5 Power-Interest Grid Classification

Table 1.6: Power-Interest Grid (Adapted from Freeman [1])

Quadrant	Stakeholders	Engagement Strategy
High Power, High Interest	Project Sponsor, Sanskrit Linguists	Manage Closely – Monthly meetings, weekly reviews
High Power, Low Interest	IGNCA, University Administration	Keep Satisfied – Quarterly reports
Low Power, High Interest	Researchers, Graduate Students	Keep Informed – Monthly newsletter, beta access
Low Power, Low Interest	Cloud Provider	Monitor – Automated alerts

1.2.6 Stakeholder Engagement Strategy

Table 1.7: Stakeholder Engagement Plan

Stakeholder	Engagement Method	Frequency
UGC and MHRD	Formal status reports, steering committee	Monthly
Sanskrit Linguists	Joint review sessions, pair validation	Weekly
NLP Engineers	Daily stand-ups, sprint planning	Daily
Researchers	User feedback surveys, beta testing	Per release
IGNCA	Quarterly review meeting	Quarterly
Graduate Students	Documentation review, usability testing	Per milestone

1.3 Experiment 3: Project Cost Estimation and Capital Budgeting

1.3.1 Cost Components Breakdown

Table 1.8: Cost Components Breakdown

Cost Category	Description	Estimated Range (INR)
Development	NLP model development, Panini grammar engine, API development	18–22 Lakhs
Data Annotation	Corpus collection, expert annotation, validation	6–8 Lakhs
Infrastructure	Cloud compute (AWS/GCP), storage, bandwidth	3–4 Lakhs
Tools & Licenses	NLP frameworks, ML libraries, IDEs	0.5–1 Lakhs
Testing & QA	Accuracy validation, performance testing	2 Lakhs
Project Management	PM tools, collaboration software	1 Lakhs
Contingency (15%)	Unforeseen expenses, risk mitigation	4–5 Lakhs
Total		**33–41 Lakhs**

1.3.2 Software Cost Estimation Models

COCOMO (Intermediate Organic Mode)

The Constructive Cost Model (COCOMO) Intermediate model [10] was applied:

Table 1.9: COCOMO Input Parameters

Parameter	Value
Project Type	Organic (relatively small team, familiar environment, but complex domain)
Estimated Project Size	25–30 KLOC (thousand lines of code)
Effort Formula	$E = a \times (KLOC)^b \times \prod_{i=1}^{n}(EM_i)$
Organic Mode Constants	$a = 3.2$, $b = 1.05$

Effort Multipliers (Selected):

- Required software reliability: High (1.15)
- Product complexity: Very High (1.30) – due to Panini rule complexity
- Analyst capability: High (0.86) – senior NLP experts required
- Programmer capability: High (0.86)
- Application experience: Low (1.13) – niche Sanskrit NLP domain
- Use of software tools: High (0.91)

- Required development schedule: Normal (1.00)

Calculated Effort:

$$E = 3.2 \times (27.5)^{1.05} \times (1.15 \times 1.30 \times 0.86 \times 0.86 \times 1.13 \times 0.91)$$

$$E \approx 3.2 \times 33.2 \times 1.12 \approx 119 \text{ Person-Months}$$

Adjusted Effort: 70–90 Person-Months

Function Point Analysis

Function Point Analysis [3] was performed:

Table 1.10: Function Point Counts

Function Type	**Count**	**Complexity Weight**
External Inputs (Sanskrit text submission)	3	High (6)
External Outputs (Parsed grammar trees)	4	High (7)
External Inquiries (API queries)	3	Medium (4)
Internal Logical Files (Panini rule database)	2	High (15)
External Interface Files (Corpus imports)	2	Medium (7)

Unadjusted Function Points (UFP) $= (3 \times 6) + (4 \times 7) + (3 \times 4) + (2 \times 15) + (2 \times 7) = 102$

Value Adjustment Factor (VAF) – Total Degree of Influence: 40

$$\text{VAF} = 0.65 + (40/100) = 1.05$$

$$\text{Adjusted Function Points} = 102 \times 1.05 = 107 \text{ FP}$$

Considering high complexity of Panini grammar, **estimated FP of 450–500** was adopted.

Expert Judgment

Expert judgment from NLP engineers and Sanskrit researchers yielded:

- Sanskrit NLP niche domain → specialist cost premium of 30–40%
- Rule-engine development inflates effort by 25%
- Data annotation requires linguist involvement → INR 8–12 per word

Consensus Estimate: INR 33–41 Lakhs

1.3.3 Capital Budgeting: Net Present Value (NPV)

Assumptions

Table 1.11: NPV Assumptions

Parameter	Value
Discount rate (cost of capital)	10% per annum
Project life	5 years (after development)
Initial investment (Year 0)	INR 37 Lakhs (midpoint)
Annual operating costs	INR 6 Lakhs

Annual Benefits:

- Academic licenses (10 institutions @ INR 1.5 Lakhs) = INR 15 Lakhs
- Research grants = INR 5 Lakhs
- API subscriptions = INR 3 Lakhs
- **Total**: INR 23 Lakhs (Year 1-2) → INR 28 Lakhs (Year 3-5)

Net annual cash flow: INR 17 Lakhs (Year 1-2), INR 22 Lakhs (Year 3-5)

NPV Calculation

$$NPV = -37 + \frac{17}{1.10} + \frac{17}{(1.10)^2} + \frac{22}{(1.10)^3} + \frac{22}{(1.10)^4} + \frac{22}{(1.10)^5}$$

$$NPV = -37 + 15.45 + 14.05 + 16.53 + 15.03 + 13.66 = 37.72 \text{ Lakhs}$$

Payback Period

Cumulative cash flows:

- Year 0: -37 Lakhs
- Year 1: -20 Lakhs

- Year 2: -3 Lakhs
- Year 3: +19 Lakhs

$$\text{Payback Period} = 2 + \frac{3}{22} = 2.14 \text{ years}$$

1.3.4 Capital Budgeting Comparison

Table 1.12: Capital Budgeting Techniques Comparison

Technique	Result	Decision
Net Present Value (NPV)	INR +37.72 Lakhs	Accept (positive)
Payback Period	2.14 years	Accept ($<$ 5 years)
Internal Rate of Return (IRR)	$\approx 42\%$	Accept ($>$ 10%)
Profitability Index (PI)	$74.72/37 = 2.02$	Accept ($>$ 1)

1.4 Experiment 4: Market Demand Analysis and Demand Planning

1.4.1 Market Demand Drivers

1. **Growing Interest in Digital Humanities**: Global market USD 4.2 billion (2025), CAGR 14.3%
2. **Government-Funded Language Preservation**: National Mission for Manuscripts – 5 million manuscripts [12]
3. **EdTech Platforms**: Sanskrit enrollment grew 40% YoY (2023-2025)
4. **UNESCO Cultural Digitization**: Memory of the World Programme
5. **AI Research in Low-Resource Languages**: Funding from DARPA, European Research Council

1.4.2 Target Market Segments

Table 1.13: Target Market Segments

Segment	Demand Level	Estimated Market Size

Universities & Research Institutes	High	120 (India) + 80 (International)
Digital Libraries	Medium	40 major collections globally
Language Learning Platforms	Medium	15–20 platforms
Government Cultural Bodies	High	8–10 (NMM, IGNCA, Sanskruti)
AI Research Labs	Medium	25–30 labs

1.4.3 Total Addressable Market (TAM) Estimation

Table 1.14: TAM Estimation by Year

Year	**Target**	**Revenue (INR)**	**Market Penetration**
Year 1	5 universities (free pilot)	0	Pilot phase
Year 2	15 paid institutions @ INR 1.5 Lakhs	22.5 Lakhs	7.5%
Year 3	35 institutions + 5 API subscribers	57.5 Lakhs	17.5%
Year 4	50 institutions + 10 API subscribers	85 Lakhs	25%
Year 5	60 institutions + 15 API subscribers + 3 libraries	114 Lakhs	30%

1.4.4 Demand Planning – Three-Phase Strategy

Table 1.15: Three-Phase Demand Planning Strategy

Phase	**Timeline**	**Activities & Targets**
Phase 1: Academic Adoption	Year 1	Free pilot to 5 leading Sanskrit universities; Collect feedback; Aim: 90% satisfaction
Phase 2: API & Research Tools	Years 2-3	Paid academic licenses (INR 1.5 Lakhs/year); Target: 35 institutions by Year 3

Phase 3: EdTech & Multilingual Expansion	Years 4-5	Integrate with 3 EdTech platforms; Expand to Pali, Prakrit; Aim: INR 114 Lakhs revenue

1.4.5 Competitive Analysis

Table 1.16: Competitive Analysis

Competitor	**Strengths**	**Weaknesses**	**Threat Level**
Sanskrit Heritage Engine (CNRS) [5]	Well-established, rule-based	Limited ML, complex setup	Medium
DCS (Cambridge) [13]	Good corpus search	No semantic parsing	Low
SanskritTagger (Open source)	Free, simple	Low accuracy (<70%)	Low
Manual linguistic analysis	High accuracy	Not scalable	Low

1.4.6 Market Differentiation Strategy

1. **Hybrid Approach**: Rule-based Paninian grammar + ML disambiguation [7]
2. **Semantic Parsing**: Extracts meaning, not just morphological tags
3. **API-First Design**: Easy integration for researchers
4. **Structured Knowledge Output**: Machine-readable triples
5. **Active Linguist Involvement**: Continuous expert validation

1.4.7 Risk-Adjusted Demand Forecast

Table 1.17: Risk-Adjusted Demand Forecast

Scenario	**Probability**	**Year 3 Revenue**	**Year 5 Revenue**	**NPV Impact**
Optimistic	25%	INR 80 Lakhs	INR 150 Lakhs	+55 Lakhs
Most Likely	50%	INR 57.5 Lakhs	INR 114 Lakhs	+37.7 Lakhs
Pessimistic	25%	INR 35 Lakhs	INR 70 Lakhs	+15 Lakhs

1.4.8 Software Process Models

The project follows a hybrid software process model combining:

- **Traditional Waterfall Model**: For Panini rule formalization (sequential, well-defined)
- **Incremental Model**: For morphological analyzer (releases v1, v2, v3)
- **Agile Methodology**: For API development and testing (2-week sprints)

1.4.9 Conclusion for Unit 1

The requirement gathering phase successfully identified 10 functional requirements and 8 non-functional requirements, prioritized using the MoSCoW method [6]. The project identification methodology confirmed technical, economic, and operational feasibility with an estimated budget of INR 33-41 Lakhs, following COCOMO and Function Point Analysis [3,10]. Stakeholder analysis [1] identified 9 key stakeholders with tailored engagement strategies. Market demand analysis projects INR 114 Lakhs in annual revenue by Year 5 with a three-phase adoption strategy.